

NASA Ingenuity Mars Helicopter Rotor Blade

Complete CAD Reconstruction Project Documentation

PROJECT OVERVIEW

This document presents a complete engineering reconstruction of the NASA Ingenuity Mars Helicopter rotor blade based entirely on **publicly available NASA data** from NASA Technical Memorandum 20240001510 (Koning et al., February 2024).

Project Type: CAD modeling exercise using published technical data
Data Source: NASA Technical Memorandum - Mars Helicopter Ingenuity Rotor Geometry
Tools Used: MATLAB (geometry processing) + SolidWorks (3D CAD)

IMPORTANT: This project uses ONLY publicly released NASA documentation. No proprietary or confidential information was accessed or used.

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1. INTRODUCTION TO NASA INGENUITY

Historic Achievement

NASA's Ingenuity Mars Helicopter achieved the **first powered, controlled flight on another planet** on April 19, 2021. This groundbreaking achievement opened new possibilities for planetary exploration.

Mission Facts

- **Maiden Flight:** April 19, 2021
- **Mission Duration:** April 2021 - January 2024 (72 flights total)
- **Total Mass:** ~1.8 kg
- **Rotor Configuration:** Coaxial counter-rotating
- **Rotor Diameter:** 1.21 m (2×0.605 m radius)
- **Operating Speed:** Up to 2,800 RPM
- **Tip Speed:** ~177 m/s
- **Tip Mach Number:** ~0.76 in Mars atmosphere
- **Flight Duration:** Up to 90 seconds per flight
- **Maximum Speed:** 10 m/s horizontal, 3 m/s vertical

The Mars Atmosphere Challenge

Flying on Mars presents extraordinary engineering challenges:

Parameter	Mars	Earth (Sea Level)	Ratio
Atmospheric Density	~0.015 kg/m ³	~1.225 kg/m ³	1:82
Temperature	~-60°C avg	+15°C avg	-
Speed of Sound	~240 m/s	~340 m/s	0.7×
Reynolds Number	~10,000	>1,000,000	1:100

Implications for Design:

- **Ultra-low density** requires very high rotational speeds
- **Low Reynolds number** reduces aerodynamic efficiency
- **Reduced speed of sound** increases Mach number effects
- **Lightweight structure** mandatory (but must withstand centrifugal loads)

Rotor Design Features

To overcome these challenges, Ingenuity's rotors feature:

- **High aspect ratio blades** for maximum lift
- **Extreme lightweight construction** (carbon fiber)
- **Custom low-Reynolds-number airfoils**
- **Strong twist distribution** (17.6° to 0°)
- **Significant chord taper** (119 mm to 7 mm)
- **Counter-rotating coaxial configuration** for stability

2. PROJECT OBJECTIVES

Primary Goal

Create a fully parametric 3D CAD model of the Ingenuity rotor blade from published NASA technical data, demonstrating a realistic aerospace engineering workflow.

Specific Objectives

1. ✓ Extract and interpret NASA-published blade geometry data
2. ✓ Process normalized airfoil coordinates (scaling, rotation)
3. ✓ Automate geometry preparation using MATLAB
4. ✓ Generate SolidWorks-compatible curve files (.sldcrv)
5. ✓ Build parametric 3D blade model via lofting

6. ✓ Document complete traceability from source data to final CAD

What This Project Demonstrates

- **Data interpretation** from aerospace technical publications
- **Automation** of repetitive geometry tasks
- **CAD modeling** using industry-standard tools
- **Engineering documentation** practices
- **Workflow** similar to real-world rotor development

3. DATA SOURCE

Primary Reference Document

NASA Technical Memorandum 20240001510

Title: Mars Helicopter Ingenuity Rotor Geometry

Authors: Witold J. F. Koning, Michelle Dominguez

Institution: NASA Ames Research Center, Moffett Field, California

Publication Date: February 2024

Availability: Public domain via NASA STI repository (ntrs.nasa.gov)

Document Contents Used

From this NASA Technical Memorandum, we extracted:

1. **Table 2:** Complete blade station geometry

- Radial positions
- Chord lengths
- Twist angles
- Airfoil assignments

2. **Appendix A:** Design airfoil coordinates

- Station 1 airfoil ($r/R = 0.0908$)
- Station 2 airfoil ($r/R = 0.2000$)
- Station 3 airfoil ($r/R = 0.2950$)

- Station 4 airfoil ($r/R = 0.3903$)
- CLF5605 airfoil (Stations 5-12)

Additional References

The NASA document cites the following development work:

- Pipenberg et al. (2019): Design and fabrication details
- Koning, Johnson, Allan (2018): Rotor performance modeling
- Balaram et al. (2018): Overall system design
- CAMRADII analysis for twist optimization

4. BLADE DESIGN STATIONS

Station Definition

The Ingenuity blade is defined at **12 discrete radial stations** rather than continuous equations. Each station specifies four parameters:

1. **r** = Radial position from rotation axis (mm)
2. **c** = Local chord length (mm)
3. **ϕ** = Local twist angle (degrees, nose-up positive)
4. **Airfoil** = Assigned cross-sectional profile

Complete Station Table

Station	r (mm)	c (mm)	ϕ (deg)	r/R	c/R	Airfoil	t/c
1	55	31	16.316	0.0909	0.0512	Station 1	Variable
2	121	85	17.616	0.2000	0.1405	Station 2	0.22
3	178	119	15.916	0.2942	0.1967	Station 3	0.10
4	236	119	12.067	0.3901	0.1967	Station 4	0.06
5	319	98	8.433	0.5273	0.1620	CLF5605	0.05
6	395	83	5.858	0.6529	0.1372	CLF5605	0.05
7	461	73	3.926	0.7620	0.1207	CLF5605	0.05
8	499	67	2.901	0.8248	0.1107	CLF5605	0.05
9	559	52	1.390	0.9240	0.0860	CLF5605	0.05
10	588	35	0.543	0.9719	0.0579	CLF5605	0.05
11	600	21	0.057	0.9917	0.0347	CLF5605	0.05
12	605	7	0.000	1.0000	0.0116	CLF5605	0.05

Note: r/R and c/R are normalized by blade radius R = 605 mm

Geometric Trends

chord Distribution:

- Peak chord at $r = 178\text{-}236$ mm (119 mm)
- Strong taper from $r = 319$ mm outward
- Tip chord only 7 mm (6% of maximum)

Twist Distribution:

- Maximum twist 17.6° at $r = 121$ mm
- Decreases monotonically to 0° at tip
- Total twist range: 17.6°

Airfoil Distribution:

- Inboard (Stations 1-4): Custom structural airfoils
- Outboard (Stations 5-12): CLF5605 aerodynamic airfoil
- Transition at $r = 319$ mm ($r/R = 0.53$)

Design Implications

1. **Inboard sections** prioritize structural strength for hub attachment
2. **Midspan sections** provide maximum thrust production
3. **Outboard sections** optimize efficiency at high speed (approaching $M = 0.76$)

5. DATA SOURCE

Primary Reference Document

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- Station 2 airfoil ($r/R = 0.2000$)
- Station 3 airfoil ($r/R = 0.2950$)
- Station 4 airfoil ($r/R = 0.3903$)
- CLF5605 airfoil (Stations 5-12)

6. BLADE DESIGN STATIONS

Station Definition

The Ingenuity blade is defined at **12 discrete radial stations** rather than continuous equations. Each station specifies four parameters:

5. **r** = Radial position from rotation axis (mm)
6. **c** = Local chord length (mm)
7. **ϕ** = Local twist angle (degrees, nose-up positive)
8. **Airfoil** = Assigned cross-sectional profile

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1	55	31	16.316	0.0909	0.0512	Station 1	Variable
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Note: r/R and c/R are normalized by blade radius $R = 605$ mm

Geometric

Trends Chord

Distribution:

- Peak chord at $r = 178\text{-}236$ mm (119 mm)
- Strong taper from $r = 319$ mm outward
- Tip chord only 7 mm (6% of maximum)

Twist Distribution:

- Maximum twist 17.6° at $r = 121$ mm
- Decreases monotonically to 0° at tip
- Total twist range: 17.6°

Airfoil Distribution:

- Inboard (Stations 1-4): Custom structural airfoils
- Outboard (Stations 5-12): CLF5605 aerodynamic airfoil
- Transition at $r = 319$ mm ($r/R = 0.53$)

Design Implications

4. **Inboard sections** prioritize structural strength for hub attachment
5. **Midspan sections** provide maximum thrust production
6. **Outboard sections** optimize efficiency at high speed (approaching $M = 0.76$)

7. AIRFOIL GEOMETRY

Airfoil Design Philosophy

Low Reynolds Number Challenge

At $Re \approx 10,000$, conventional airfoils suffer from:

- **Premature laminar separation**
- **Large separation bubbles**
- **Reduced lift capability**
- **Increased drag**

CLF5605 Design Features

The primary outboard airfoil (CLF5605) was specifically designed for low-Re Mars conditions:

- **Thin profile:** $t/c = 5\%$ (minimum for structural frequency requirements)
- **Laminar flow optimization:** Extended favorable pressure gradient
- **Increased camber:** Higher lift coefficients than baseline
- **Heritage design:** Based on proven AeroVironment low-Re airfoils

Coordinate System Convention

All airfoils use normalized coordinates:

- **x/c :** Chordwise position (0 = leading edge, 1 = trailing edge)
- **y/c :** Normal to chord (+ = upper surface, - = lower surface)

Coordinate ordering:

1. Start: Trailing edge ($x/c = 1.0$, $y/c = 0.0$)
2. Progress along **upper surface** to leading edge
3. Return along **lower surface** to trailing edge

4. Close loop at trailing edge

Manufacturing Modifications

Published Design: Infinitely sharp trailing edge

As-Built Modification:

- Linear thickness increase near trailing edge
- Final TE thickness: ~ 0.5 mm across span
- Maintains aerodynamic performance
- Ensures manufacturability

8. COMPLETE RAW AIRFOIL DATA

The following sections contain **ALL raw normalized coordinate data** exactly as provided in NASA TM-20240001510, Appendix A.

Station 1 Airfoil

Location: $r = 55$ mm ($r/R = 0.0909$)

Chord: $c = 31$ mm

Twist: $\varphi = 16.316^\circ$

Purpose: Root/hub attachment

Station 2 Airfoil

Location: $r = 121$ mm ($r/R = 0.2000$)

Chord: $c = 85$ mm

Twist: $\varphi = 17.616^\circ$ (maximum twist)

Station 3 Airfoil

Location: $r = 178$ mm ($r/R = 0.2950$)

Chord: $c = 119$ mm (maximum chord)

Twist: $\varphi = 15.916^\circ$

Station 4 Airfoil

Location: $r = 236 \text{ mm}$ ($r/R = 0.3903$)

Chord: $c = 119 \text{ mm}$

Twist: $\phi = 12.067^\circ$

CLF5605 Airfoil (Stations 5-12)

Used at: $r = 319, 395, 461, 499, 559, 588, 600, 605 \text{ mm}$

Thickness: $t/c = 5.0\%$ (constant)

Type: Low-Reynolds-number laminar flow airfoil

(ALL DATA REGARDING IN EXCEL FILE)

9. MATLAB GEOMETRY PROCESSING

Processing Requirements

For each of the 12 blade stations, we need to:

- A. Read raw normalized airfoil coordinates (x/c , y/c)
- B. Remove any header lines or non-numeric data
- C. **Scale** coordinates by local chord length: $x = (x/c) \times c$, $y = (y/c) \times c$
- D. **Rotate** airfoil by local twist angle ϕ
- E. **Translate** to correct radial position r
- F. Ensure trailing edge closure
- G. Export as SolidWorks-compatible .sldcrv file

10. SOLIDWORKS CAD MODELING

Import Procedure

Step 1: create reference planes

- Create 12 planes perpendicular to rotation axis
- Position each at correct radial distance r

Step 2: Import Station Curves Step : create reference planes

- Use Insert → Curve → Curve Through XYZ Points
- Select corresponding .sldcrv file
- Verify curve lies on correct plane

Step 3: Verify Geometry

- Check chord lengths
- Verify twist progression
- Ensure proper closure at trailing edges

Blade Surface Generation

Method: Lofted Boss/Base

Settings:

- Guide curves: None (stations define shape)
- Start/End constraints: Normal to path
- Merge tangent faces: Yes

Process:

- H. Select all 12 station curves in order
- I. Create loft with smooth transitions
- J. Verify no surface irregularities
- K. Check edge quality at root and tip

Model Quality Checks

- ✓ Chord distribution matches Table 2
- ✓ Twist variation smooth and continuous

- ✓ No surface defects or kinks
- ✓ Proper closure at trailing edges
- ✓ Blade mass/volume reasonable

Final Model

Features Geometry:

- Single continuous surface
- 12 defining sections
- Smooth lofted transitions

Properties:

- Total blade length: 605 mm
- Root chord: 31 mm
- Tip chord: 7 mm
- Surface area: $\sim 0.17 \text{ m}^2$ (per NASA)

11. RESULTS & CONCLUSIONS

Project Achievements

- ✓ **Complete geometric reconstruction** from published NASA data
- ✓ **Full traceability** from source document to final CAD
- ✓ **Automated workflow** minimizing manual errors
- ✓ **Parametric model** suitable for further analysis
- ✓ **Documentation** following aerospace engineering practices

Model Validation

The reconstructed blade geometry accurately reproduces:

- All 12 station positions and orientations
- Chord taper (119 mm $\rightarrow$ 7 mm)
- Twist distribution ($17.6^\circ \rightarrow 0^\circ$)
- Airfoil transitions (4 custom + CLF5605)
- Overall blade dimensions and proportions

Engineering

Learnings Key

Insights:

1. Number of stations $\neq$ number of airfoils (12 stations, 5 airfoils)
2. MATLAB automation greatly improves accuracy and speed
3. Separating data processing from CAD improves workflow
4. Published aerospace data is well-suited for reconstruction

Design Observations:

- Extreme taper required for Mars conditions
- Structural sections at root, aerodynamic sections outboard
- Twist optimized for efficiency across operating range
- Low thickness ratio (5%) driven by frequency requirements

Applications of This Model

The completed CAD model can be used for:

- ✓ **CFD Analysis:** Aerodynamic performance prediction
- ✓ **FEA Analysis:** Structural loads, vibration modes
- ✓ **Manufacturing:** 3D printing, machining toolpaths
- ✓ **Education:** Understanding Mars helicopter design
- ✓ **Research:** Low-Re rotor aerodynamics studies

Limitations

This reconstruction does NOT include:

- Detailed hub geometry (not published by NASA)
- Root attachment features
- Manufacturing tolerances
- Surface finish specifications
- Internal structure/reinforcement

This is a geometry-only model based on outer mold line data.

REFERENCES

Primary Source

[1] Koning, W. J. F., and Dominguez, M., "Mars Helicopter Ingenuity Rotor Geometry," NASA Technical Memorandum 20240001510, NASA Ames Research Center, February 2024.

Supporting References

[2] Pipenberg, B. T., Keennon, M., Tyler, J., et al., "Design and Fabrication of the Mars Helicopter Rotor, Airframe, and Landing Gear Systems," AIAA SciTech Forum, AIAA Paper 2019-0620, January 2019.

[3] Koning, W. J. F., Johnson, W., and Grip, H. F., "Improved Mars Helicopter Aerodynamic Rotor Model for Comprehensive Analyses," AIAA Journal, Vol. 57, No. 9, September 2019.

[4] Balaram, J., et al., "Mars Helicopter Technology Demonstrator," AIAA SciTech Forum, AIAA Paper 2018- 0023